

NAKHONPATHOM, Thailand

WMO: 484510

Lat: 14.0117N

Lon: 99.9700E

Elev: 10

StdP: 101.21

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 16.8	(c) 18.3	(d) 12.1	(e) 8.8	(f) 20.1	(g) 14.2	(h) 10.1	(i) 21.9	(j) 4.5	(k) 25.7	(l) 4.2	(m) 25.8	(n) 0.5	(o) N/A	(p) 0.323

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 9.6	(c) 37.3	(d) 26.6	(e) 36.3	(f) 26.6	(g) 35.5	(h) 26.6	(i) 28.2	(j) 33.3	(k) 27.9	(l) 33.0	(m) 27.6	(n) 32.6	(o) 1.7	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.2	23.0	29.9	26.7	22.4	29.6	26.5	22.0	29.4	90.9	33.7	89.3	33.2	88.0	32.9	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
4.8	4.0	3.3	DB	14.3	38.7	2.2	1.3	12.8	39.7	11.5	40.4	10.3	41.1	8.7	42.0	
			WB	13.2	29.3	2.5	0.7	11.5	29.8	10.0	30.2	8.6	30.6	6.9	31.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.5	25.3	27.7	29.4	30.8	30.6	29.7	29.2	29.1	28.9	28.3	27.2	25.3
	DBStd	2.37	2.42	1.88	2.00	1.71	1.53	1.02	1.06	0.93	1.01	1.08	1.55	2.32
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6739	476	496	601	624	638	591	594	593	567	567	517	476
	CDD18.3	3697	217	262	343	374	379	341	336	335	317	309	267	217
	CDH23.3	41696	1959	2858	4219	4924	4804	4080	3859	3782	3468	3292	2596	1852
	CDH26.7	18527	735	1308	2103	2687	2447	1826	1613	1555	1398	1252	955	649
Wind	WSAvg	0.9	0.7	0.8	1.1	1.1	1.0	0.9	0.9	1.0	0.9	0.7	0.8	0.9
Precipitation	PrecAvg	1071	7	14	36	66	137	101	108	113	232	208	43	6
	PrecMax	1397	53	92	175	233	276	192	184	184	434	366	132	40
	PrecMin	743	0	0	0	1	17	12	50	52	104	29	0	0
	PrecStd	153	11	21	40	54	59	42	40	34	85	81	41	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	35.4	37.3	38.9	38.7	36.5	35.7	35.2	35.1	34.5	34.0	33.9
		MCWB	25.5	25.7	24.6	26.4	27.2	27.2	26.6	26.7	27.1	26.8	26.8	25.5
	2%	DB	32.8	34.6	36.1	37.6	37.2	35.2	34.4	34.2	34.2	33.5	33.0	32.8
		MCWB	24.9	25.5	25.4	26.7	27.3	27.2	26.6	26.7	26.9	26.7	26.1	25.3
	5%	DB	31.9	33.7	35.3	36.6	36.0	34.2	33.5	33.4	33.3	32.6	32.2	31.7
		MCWB	24.6	25.2	25.8	26.7	27.2	27.0	26.4	26.6	26.8	26.6	25.8	25.0
	10%	DB	30.7	32.8	34.4	35.5	34.9	33.2	32.6	32.5	32.4	31.8	31.2	30.2
		MCWB	24.2	25.1	25.8	26.8	27.1	26.9	26.3	26.4	26.6	26.4	25.5	24.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.7	27.3	27.7	28.7	29.0	28.2	27.8	27.9	28.0	27.9	27.7	26.7
		MCDB	31.6	33.1	33.5	33.7	33.9	33.4	32.8	32.4	32.8	31.7	31.6	31.1
	2%	WB	26.1	26.7	27.2	28.1	28.2	27.7	27.2	27.4	27.6	27.5	27.2	26.1
		MCDB	30.8	32.2	32.7	33.4	33.5	33.0	32.2	32.0	32.2	31.3	31.3	30.8
	5%	WB	25.7	26.3	26.9	27.7	27.9	27.5	26.9	27.0	27.2	27.1	26.7	25.5
		MCDB	30.0	31.4	32.1	33.0	33.2	32.5	31.9	31.7	31.7	31.0	30.6	29.9
	10%	WB	25.1	25.9	26.6	27.4	27.5	27.1	26.6	26.7	26.9	26.8	26.2	25.0
		MCDB	28.9	30.6	31.4	32.5	32.8	31.9	31.1	31.0	31.2	30.6	29.6	29.0
Mean Daily Temperature Range	5% DB	MDBR	10.0	10.3	9.7	9.6	8.8	7.8	7.4	7.5	7.5	7.1	7.8	9.1
		MCDBR	11.3	11.0	10.9	10.7	10.3	9.0	8.6	8.6	8.8	8.4	9.0	10.4
		MCWBR	4.0	3.1	2.8	2.7	2.6	2.7	2.6	2.6	2.8	2.7	2.9	3.6
	5% WB	MCDBR	9.6	9.9	9.8	9.9	9.3	8.2	8.0	7.9	8.2	7.7	8.0	9.5
MCWBR		3.5	3.0	2.9	2.8	2.9	2.7	2.6	2.7	2.9	2.7	3.0	3.5	
Clear-Sky Solar Irradiance	taub		0.540	0.604	0.642	0.570	0.489	0.450	0.443	0.444	0.469	0.579	0.527	0.517
	taud		1.949	1.823	1.764	1.980	2.265	2.373	2.394	2.378	2.297	1.946	2.061	2.031
	Ebn at Noon		758	729	714	766	817	840	848	857	840	740	765	766
	Edh at Noon		182	214	232	187	138	122	120	124	134	187	162	164
All-Sky Solar Radiation	RadAvg		4.88	5.34	5.56	6.02	5.45	5.07	4.77	4.77	4.82	4.64	4.77	4.73
	RadStd		0.29	0.23	0.42	0.36	0.44	0.35	0.32	0.34	0.31	0.29	0.32	0.22

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (29 neighbors)	N/A	N/A	N/A	N/A	+0.25	+0.27	N/A	N/A	+86	+86